

ORIGINAL ARTICLE

Clonal Hematopoiesis of Indeterminate Potential Proteomic Risk Score for Predicting Clinical Outcome in Patients With Heart Failure With Preserved Ejection Fraction

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BACKGROUND: Clonal hematopoiesis of indeterminate potential (CHIP) is common in heart failure with preserved ejection fraction (HFpEF). However, the proteomic signatures linked to CHIP driver mutations and their prognostic implications in HFpEF remain poorly defined. We aimed to identify plasma protein signatures associated with CHIP driver mutations and to develop and externally validate a CHIP-proteomic risk score (CHIP-ProtRS) for predicting clinical outcomes in HFpEF.

METHODS: The derivation cohort consisted of 118 TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) participants with available CHIP sequencing and proteomic profiling using the SomaScan platform. Associations between CHIP mutations and circulating proteins were assessed using gene-specific linear regression adjusted for age, with correction for multiple comparisons. Significant proteins were used to derive gene-specific proteomic classifiers using nested cross-validated least absolute shrinkage and selection operator logistic regression. A composite CHIP-ProtRS was generated using least absolute shrinkage and selection operator Cox regression and validated in a composite of 3 independent HFpEF cohorts (n=654). The primary end point was a composite of death and HF hospitalization.

RESULTS: In TOPCAT, 30.5% of tested participants carried ≥ 1 CHIP mutation. Significant protein associations were identified for *TET2* (7 proteins), *DNMT3A* (10), and *PPM1D* (5), all of which were enriched in immune and inflammatory pathways. Gene-specific least absolute shrinkage and selection operator models showed good discrimination for CHIP mutation (area under the curves, 0.752–0.898). The primary end point occurred in 28.0% of the derivation cohort and 34.7% of the validation cohorts. In the Cox-least absolute shrinkage and selection operator combination, only the *TET2*-derived proteomic signature retained a nonzero coefficient in the CHIP-ProtRS. The CHIP-ProtRS was independently associated with the primary end point after adjustment for age, sex, Meta-Analysis Global Group in Chronic Heart Failure score, NT-proBNP (N-terminal pro-B-type natriuretic peptide), and atrial fibrillation in the derivation (adjusted hazard ratio, 1.42 [95% CI, 1.01–2.00]) and validation cohorts (adjusted hazard ratio, 1.16 [95% CI, 1.01–1.32]).

CONCLUSIONS: CHIP is associated with alterations in plasma concentrations of immune/inflammatory proteins. A CHIP-ProtRS is independently associated with adverse outcomes in HFpEF, supporting its potential role as a biomarker of risk and underlying pathobiology.

Key Words: clonal hematopoiesis ■ diastolic heart failure ■ prognosis ■ proteomics ■ risk assessment

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WHAT IS NEW?

- Clonal hematopoiesis of indeterminate potential (CHIP) is increasingly recognized as a risk factor for adverse outcomes in heart failure with preserved ejection fraction (HFpEF), but the downstream biological mechanisms mediating this risk have remained unclear.
- Using large-scale plasma proteomic profiling, this study identifies distinct circulating protein signatures associated with major CHIP driver mutations (*TET2*, *DNMT3A*, and *PPM1D*) in patients with HFpEF.
- We developed and externally validated a CHIP-related proteomic risk score that consistently stratifies the risk of death and heart failure hospitalization across multiple independent HFpEF cohorts, after adjustment for established clinical risk markers.

WHAT ARE THE CLINICAL IMPLICATIONS?

- CHIP-associated alterations in circulating proteins capture inflammatory and immune pathways relevant to HFpEF progression and are independently associated with adverse clinical outcomes after adjustment for the Meta-Analysis Global Group in Chronic Heart Failure risk score, atrial fibrillation, and NT-proBNP (N-terminal pro-B-type natriuretic peptide).
- The CHIP-proteomic risk score enables identification of patients with HFpEF at higher risk of adverse outcomes even in the absence of genetic sequencing data, supporting the potential role of proteomics as a surrogate for CHIP-related biological activity.
- These findings provide biological insight into CHIP-related pathobiology in HFpEF and highlight CHIP-related pathways as potential targets for future mechanistic and therapeutic studies.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
CH	clonal hematopoiesis
CHIP	clonal hematopoiesis of indeterminate potential
CHIP-ProtRS	clonal hematopoiesis of indeterminate potential proteomic risk score
HF	heart failure
HFpEF	heart failure with preserved ejection fraction
HR	hazard ratio
IQR	interquartile range
LASSO	least absolute shrinkage and selection operator
MAGGIC	Meta-Analysis Global Group in Chronic Heart Failure

Clonal hematopoiesis (CH) refers to the selective over-expansion of a somatically mutated hematopoietic clone that becomes overrepresented in peripheral blood. Despite the potential for malignant transformation with further mutational events, most carriers never develop clinically apparent hematologic disease, and this condition is termed CH of indeterminate potential (CHIP).¹ CHIP is associated with atherosclerosis and adverse cardiovascular outcomes.^{2,3} Subsequent studies have linked CHIP to multiple cardiovascular conditions, including valvular heart disease and heart failure (HF).^{4–9} HF with preserved ejection fraction (HFpEF) has distinct pathophysiologic mechanisms. Dysregulated immune function and inflammation have been implicated as central drivers of HFpEF and are associated with adverse outcomes.¹⁰ Recent studies have implicated CHIP in the pathophysiology of HFpEF.^{7,8} CHIP is associated with a higher risk of incident HFpEF⁹ and worse clinical outcomes in those with prevalent disease.⁷ However, the specific biologic processes through which CHIP may contribute to adverse outcomes in patients with HFpEF remain unknown.

Circulating proteins capture dynamic biological activity and the downstream effects of genetic and environmental influences. Modern high-throughput proteomic platforms enable the simultaneous measurement of thousands of plasma proteins, providing a broad molecular view of an individual's physiological state. Such protein signatures have recently been shown to enhance cardiovascular risk stratification.^{11,12} Proteomic profiling enables direct interrogation of human pathophysiology, providing access to molecular processes that cannot be fully captured in animal models, and thus offers a powerful approach to delineate CHIP-related biologic abnormalities in humans. We aimed to examine CHIP-related alterations in circulating proteins in HFpEF and to assess whether a CHIP-proteomic risk score (CHIP-ProtRS) is associated with clinical outcomes across HFpEF cohorts.

METHODS

The data supporting the findings of this study may be made available for collaborative research upon execution of appropriate data-sharing agreements.

Study Participants: TOPCAT Derivation Cohort

We used plasma samples from the TOPCAT trial (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist), obtained through the US National Heart, Lung, and Blood Institute BioLINCC data and biospecimen (NHLBI BioLINCC) resource. The study design and primary results of the TOPCAT trial have been previously reported.¹³ Briefly, TOPCAT was a multinational, randomized, double-blind study comparing spironolactone with placebo and enrolled 3445 participants with HFpEF between 2006 and 2012.^{13,14} Eligible participants were ≥50 years of age, had a left ventricular ejection

fraction of at least 45%, presented with both symptoms and signs of HF, and had either a HF-related hospitalization within the preceding 12 months or an elevated natriuretic peptide concentration (BNP ≥ 100 pg/mL or NT-proBNP [N-terminal pro-B-type natriuretic peptide] ≥ 360 pg/mL) within 60 days before randomization.^{13,14} Among the 1765 participants from the American study sites, 218 had proteomic data and 118 of these also had independently performed CHIP mutation sequencing; these 118 participants comprised the derivation cohort—a convenience subset defined by the intersection of 2 independently generated biospecimen data sets, consistent with prior TOPCAT biomarker substudies.^{7,15} TOPCAT was conducted with the approval of local institutional review boards. The study participants also provided informed consent.

The raw data of this study are not publicly available for purposes of reproducing the results or replicating the procedures. These data may be available subject to the establishment of appropriate data-sharing agreements and regulatory approvals. The parent TOPCAT trial data are available through the US National Institutes of Health BioLINCC repository.

External Validation Cohorts

We used 3 external HFpEF cohorts with available proteomic data for validation: the PHFS (Penn Heart Failure Study), the WashU registry (Washington University Heart Failure), and the PEOPLE study (Prospective Evaluation of Outcome in Patients With Heart Failure With Preserved Left Ventricular Ejection Fraction). In the external validation cohorts, HFpEF was defined according to recent guideline-based criteria, including left ventricular ejection fraction $\geq 50\%$, clinical symptoms of HF, and evidence of structural heart disease.¹⁶ The PHFS is a prospective cohort that enrolled patients receiving care at 3 US HF specialty centers (the University of Pennsylvania, Case Western Reserve University, and the University of Wisconsin) between 2003 and 2011.¹⁷ Participants were eligible if they were evaluated in an outpatient HF specialty clinic at one of these centers for diagnostic assessment or management of HF and if they had an anticipated life expectancy of at least 6 months. Among 2379 PHFS participants, 269 met criteria for HFpEF. Of these, 253 individuals had available plasma samples for proteomic profiling, and 251 participants with available follow-up data were included in the final analysis. The WashU HF registry is a prospective database of individuals with clinically diagnosed HF who are evaluated at Washington University School of Medicine.¹⁸ Among 1090 enrolled patients, 243 had HFpEF; 239 of them had available plasma samples for proteomic profiling, and 238 with follow-up information comprised the final analytic sample. The PEOPLE study is a prospective, multicenter cohort from New Zealand that includes clinically adjudicated HF patients and community-based healthy controls, designed to evaluate ethnic differences in HF characteristics and outcomes.¹⁹ Among 174 eligible HFpEF participants, 173 had available plasma samples for proteomic profiling and were included in the final cohort. Of the 662 participants across the 3 cohorts, we excluded 8 individuals with samples failing proteomic data quality control, resulting in a final external validation sample of 654 participants. The PHFS, WashU, and PEOPLE studies received appropriate ethics approvals, and all participants provided written informed consent, including permission for longitudinal data collection.

Proteomic Profiling

All plasma samples were profiled using the SomaScan platform (v4.0 or higher; SomaLogic Inc., Boulder, CO), a high-throughput proteomic technology based on modified aptamer binding. This assay employs chemically altered slow-off-rate aptamers (SOMAmer reagents) to selectively capture and quantify proteins, with signal intensity reported as relative fluorescent units proportional to protein abundance. The version used for derivation in this study comprised 5281 SOMAmer reagents targeting 5154 distinct proteins. Because individual protein measurements can exhibit a nonnormal distribution, all proteomic measurements were log-transformed before analysis to improve normalization and reduce skewness.

Assessment of CHIP

Targeted sequencing for CH mutations was performed in TOPCAT participants using a hybridization capture-based targeted sequencing panel covering 24 genes predicted to encompass $>95\%$ of CHIP mutations observed in adults (Vanderbilt VANTAGE). This panel includes the entire coding region (or curated regions involved in CHIP) of the following genes: *ASXL1*, *ASXL2*, *BRCC3*, *CBL*, *DNMT3A*, *ETNK1*, *GNAS*, *GNB1*, *IDH1*, *IDH2*, *JAK2*, *KIT*, *KRAS*, *MPL*, *NRAS*, *PPM1D*, *SETBP1*, *SF3B1*, *SRSF2*, *TET2*, *TP53*, *U2AF1*, *ZBTB33*, and *ZNF318*. An average read depth of 400 $\times$ allowed for the detection of variant allele frequencies as low as 1%. Raw data were aligned with BWA-MEM, and aligned sequencing files were analyzed with Mutect2,²⁰ followed by filtering of identified variants to include only those on a white list of variants that have also been described within 72 driver genes based on established variant calling from the myeloid malignancy field.²¹ ANNOVAR²² was then used to annotate all putative variants for downstream filtering. All variants were then manually adjudicated to exclude artifacts, including strand bias, read quality, and clustered events.

Association Between Plasma Proteins and CHIP Mutation

We evaluated the association between somatic CHIP driver mutations and circulating protein levels. For each protein-gene pair, we fit an independent linear regression model in which the log-transformed protein level was regressed on gene-specific CHIP carrier status, with age included as a covariate. Each model yielded a standardized β -coefficient (classical standardized regression coefficient, rescaled by the SD ratio of exposure and outcome), its 95% CI, and the corresponding *P* value. Given the covariation in plasma proteins, correction for multiple comparisons was performed using a principal component analysis-based approach, defining the effective number of tests as the number of principal components required to explain 95% of the variance, as previously described.²³ Statistical significance was assessed using a Bonferroni-corrected threshold based on this number.

Pathway Enrichment Analysis

For pathway analyses, we focused on CHIP driver genes with sufficient numbers of mutation carriers. For each of these genes, we selected proteins that were significantly associated

with gene-specific CHIP carrier status after PCA-based multiple testing correction ($P < 0.05$) and extracted the corresponding gene symbols. Gene symbols were mapped to Entrez identifiers and used as input for Reactome pathway enrichment (ReactomePA package in R), with significance defined as a false discovery rate-adjusted $P < 0.05$; results were summarized using standard visualization functions. To incorporate directionality, gene-level Z scores were calculated from standardized regression coefficients, and pathway-level directionality was summarized by aggregating the predominant direction of gene-level effects within each pathway.

Outcomes

In the derivation cohort, the primary end point was defined as a composite of cardiac death and HF hospitalization. In the external validation cohorts, cardiac-specific mortality was not available; therefore, the primary end point was defined as a composite of all-cause death and HF hospitalization. Secondary outcomes included individual components of the primary end point as well as major adverse cardiovascular events (defined as the composite of cardiac death, HF hospitalization, myocardial infarction, and ischemic stroke), depending on data availability in each cohort.

Generation and External Validation of CHIP-ProtRS

The CHIP-ProtRS was developed in 2 steps. First, plasma proteins associated with individual CHIP driver genes were identified using linear regression with multiple testing correction. Among the 11 genes assessed, those with ≥ 8 mutation carriers were used to derive gene-specific proteomic signatures because smaller case counts led to unstable convergence of penalized regression models. For each gene, a logistic least absolute shrinkage and selection operator (LASSO; $\alpha = 1$) regression model was trained using nested cross-validation. In the outer loop, data were split into 5 folds, with 1 fold held out for testing and the remaining 4 used for training, and this procedure was repeated 10x. Within each training set, 5-fold cross-validation was used to select the penalty parameter that maximized the mean cross-validated area under the curve. Inverse-frequency weights were applied to mutation carriers and noncarriers to account for class imbalance. These models generated gene-specific predicted mutation probabilities, which were then refit in the full derivation cohort to obtain coefficients for external application. In the second step, the gene-specific proteomic signatures were entered into a Cox LASSO regression model to predict the primary end point. The optimal penalty parameter was selected by cross-validation, and the resulting linear predictor was defined as the CHIP-ProtRS. In the external cohorts, gene-specific proteomic scores were obtained by applying the derivation-cohort LASSO coefficients to each participant's protein measurements, and the CHIP-ProtRS was then computed using the derivation-cohort Cox-LASSO coefficients.

Statistical Methods

Continuous variables are presented as mean $\pm$ SD or median (interquartile range [IQR]), depending on whether normality is satisfied by the Shapiro-Wilk test. Categorical variables

are shown as absolute counts (percentages). The categorical variables were compared using a χ^2 test and the continuous variables using Student *t* test. Associations between the CHIP-ProtRS and clinical outcomes were evaluated using Cox proportional hazards models adjusted for age, sex, Meta-Analysis Global Group in Chronic Heart Failure (MAGGIC) risk score, NT-proBNP, and atrial fibrillation (AF). Results are presented as hazard ratios (HRs) and 95% CIs for the primary and secondary outcomes per 1-SD increase in the CHIP-ProtRS. Incremental model performance after addition of the CHIP-ProtRS to the baseline clinical model was assessed using the likelihood ratio test, Harrell C-index, integrated discrimination improvement, and continuous net reclassification improvement. For survival analyses, Kaplan-Meier curves were used to compare event rates according to the CHIP-ProtRS, with differences evaluated using the log-rank test. Several sensitivity analyses were performed to evaluate the robustness of the CHIP-ProtRS. First, additional adjustment for CHIP carrier status in the multivariable Cox model was performed to determine whether CHIP-ProtRS captures prognostic information beyond mutation status itself. Second, gene-specific mutation status (*DNMT3A*, *TET2*, and *PPM1D* individually) was added as covariates. Third, body mass index was included as a separate covariate, beyond its contribution to the MAGGIC risk score. Fourth, patients with mutations among the 3 modeled genes were excluded. Finally, to assess the influence of covariate selection in the protein-mutation association step, the entire score derivation pipeline was repeated with sex, history of spironolactone, body mass index, or estimated glomerular filtration rate substituted for age in the linear regression models, followed by re-derivation of gene-specific LASSO classifiers and the composite CHIP-ProtRS. All analyses were conducted using MATLAB Statistics R2021b (The MathWorks) and R, version 4.4.2, and statistical significance was defined as a 2-sided $P < 0.05$.

RESULTS

Baseline Characteristics of the Study Participants

Baseline characteristics of the study cohorts are summarized in Table 1. The median follow-up duration was 899 days (IQR, 513–1370) in the derivation cohort and 730 days (IQR, 449–862) in the external validation cohorts. The PHFS cohort had a median follow-up period of 1139 days (IQR, 399–1855). Participants in the derivation cohort were older (73.0 versus 65.7 years; $P < 0.001$) and had higher body mass index than participants in the external validation cohorts. The prevalence of several comorbidities—including hypertension, diabetes, and chronic kidney disease—was significantly higher in the derivation cohort. Sex distribution and proportion of AF were similar across cohorts. Higher use of renin-angiotensin system inhibitors, β -blockers, loop diuretics, thiazide-like diuretics, and spironolactone was observed in the derivation cohort. NT-proBNP levels were also higher in the derivation cohort. However, New York Heart Association functional class did not differ between

Table 1. Baseline Characteristics of Study Participants

	Derivation cohort (TOPCAT; n=118)	Validation cohorts (PHFS, WashU, PEOPLE; n=654)	P value
Age, y	73.0 (65.0–80.0)	65.7 (55.1–76.8)	<0.001
Male sex	61 (51.7)	322 (49.2)	0.695
Systolic BP, mm Hg	127.0 (118.0–137.0)	124.0 (111.0–140.0)	0.417
Diastolic BP, mm Hg	70.0 (60.0–78.0)	71.0 (64.0–79.0)	0.109
BMI, kg/m ²	32.8 (28.7–37.6)	30.6 (25.6–37.1)	0.005
Comorbidities, n (%)			
Hypertension	113 (95.8)	442 (67.6)	<0.001
Diabetes	53 (44.9)	208 (31.9)	0.008
Coronary revascularization	46 (39.0)	166 (25.4)	0.003
Chronic kidney disease	51 (43.2)	137 (20.9)	<0.001
Atrial fibrillation	56 (47.5)	273 (42.9)	0.410
Medications, n (%)			
RAS inhibitors	94 (79.7)	458 (70.0)	0.043
β-Blockers	98 (83.1)	477 (72.9)	0.027
Loop diuretics	107 (90.7)	457 (69.9)	<0.001
Thiazide-like diuretics	47 (39.8)	97 (14.8)	<0.001
Spironolactone	59 (50.0)	115 (17.6)	<0.001
NYHA functional class, n (%)			
I–II	71 (60.2)	421 (64.4)	0.441
III–IV	47 (39.8)	233 (35.6)	
Laboratory findings			
eGFR, mL/min per 1.73 m ²	61.7 (48.9–76.3)	55.2 (41.5–72.4)	0.002
NT-proBNP, pg/mL	924.1 (792.5–1418.8)	427.4 (36.6–771.2)	<0.001
MAGGIC risk score	17.0 (12.0–21.0)	16.0 (10.0–23.0)	0.703

Data are presented as number (%) or mean ± SD, as appropriate. BMI indicates body mass index; BP, blood pressure; eGFR, estimated glomerular filtration rate; MAGGIC, Meta-Analysis Global Group in Chronic Heart Failure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; PEOPLE, Prospective Evaluation of Outcome in Patients With Heart Failure With Preserved Left Ventricular Ejection Fraction; PHFS, Penn Heart Failure Study; RAS, renin-angiotensin system; TOPCAT, Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist; and WashU, Washington University Heart Failure.

groups, and the median MAGGIC risk score was similar (17.0 versus 16.0; $P=0.703$). In addition, there was no difference in baseline characteristics between TOPCAT participants enrolled at sites in the Americas according to the availability of CHIP and proteomic data (Table S1).

Baseline Characteristics and Mutation Profiles According to CHIP Status

Baseline characteristics according to CHIP mutation status in the derivation cohort are summarized in Table S2. Among the 118 participants with available CHIP data, 36 individuals (30.5%) had at least 1 CHIP mutation. Participants with at least 1 CHIP mutation were older than those without mutations (77.0 versus 71.0 years; $P=0.040$). However, all other variables—including vital signs, body mass index, comorbidities, medication use, New York Heart Association functional class, estimated glomerular filtration rate, and NT-proBNP levels—were comparable between the 2 groups. Regarding mutation patterns, the mean number of mutations per individual was 1.4 ± 0.6 . The 5 most common mutations were *TET2*

(30.6%), *DNMT3A* (30.6%), *PPM1D* (25.0%), *ASXL1* (19.4%), and *TP53* (11.1%) (Figure 1). When participants were stratified by the occurrence of the primary end point occurrence, no significant differences in the distribution of CHIP-associated gene variants were observed between the 2 groups (Table 2).

Proteomic Associations of CHIP Mutations and Pathway Enrichment Analyses

We performed linear regression analyses for each CHIP mutation gene and each individual circulating protein. We identified 7 significant proteins for *TET2*, 10 for *DNMT3A*, 5 for *PPM1D*, and 6 for *ASXL1* after multiple-comparison adjustment (Table 3). Volcano plots illustrating the associations between each gene and protein levels are shown in Figure 2. *TP53* and several other genes (*TP53* and *SRSF2*: 30 proteins; *SF3B1*: 40 proteins; *KIT*: 43 proteins; *JAK2*: 465 proteins) showed a disproportionately large number of significant proteins compared with the 4 most common genes. However, because the number of mutation carriers for these genes was small, these

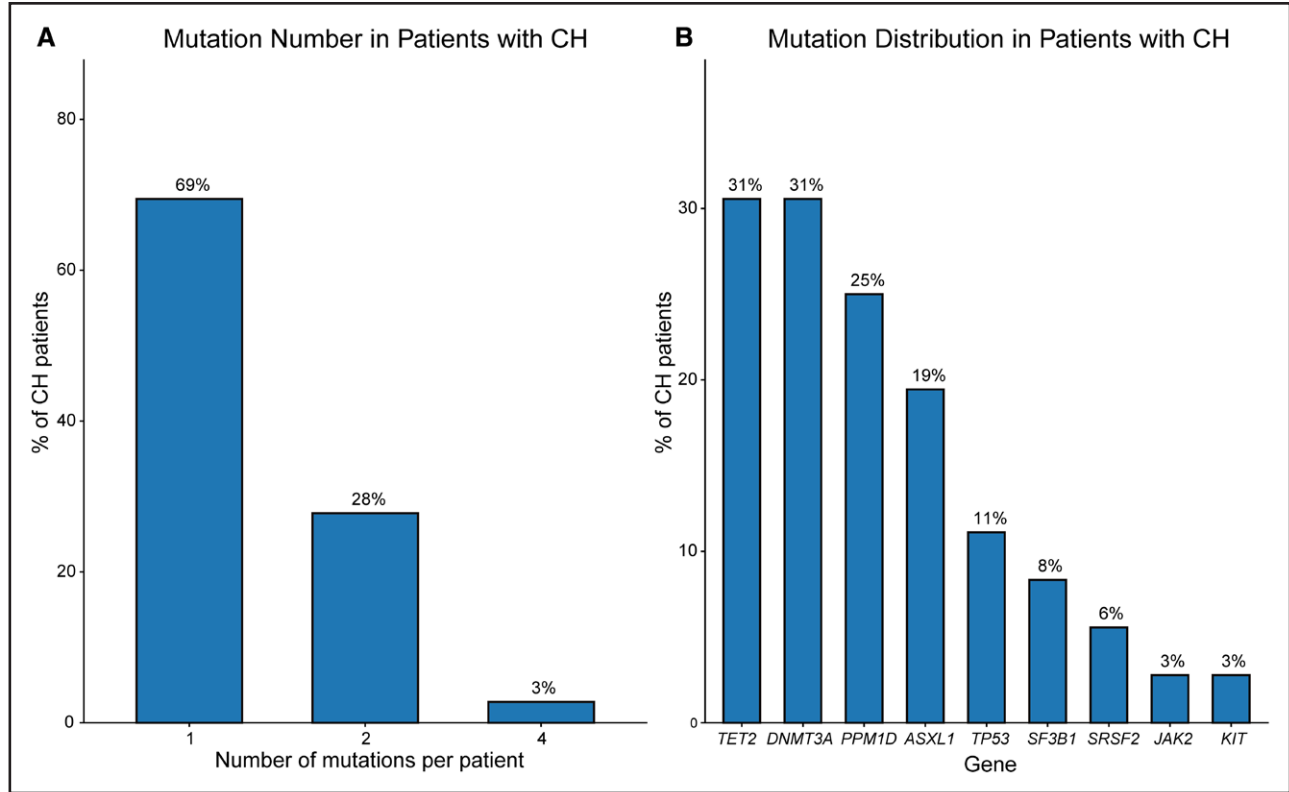


Figure 1. Mutation characteristics in patients with clonal hematopoiesis (CH): proportion of patients with CH (A) and distribution of CH variations (B) among patients in the derivation cohort.

results might have been overfit and were therefore not included in downstream analyses.

Pathway enrichment results for *TET2* and *PPM1D* are presented in Figure 3. *DNMT3A*-associated proteins did not yield any pathway meeting the false discovery rate <0.05 threshold. *TET2*-associated proteins were enriched in interleukin-mediated signaling and inflammasome-related pathways, with IL-4 and IL-13 signaling and signaling by interleukins showing a predicted activation pattern. *PPM1D*-associated proteins were enriched in neuronal and neurotransmitter-related pathways.

CHIP-ProtRS

Among CHIP mutation genes, we focused on *TET2*, *DNMT3A*, and *PPM1D*, for which at least 8 mutation carriers were available. For each of these genes, we trained a gene-specific proteomic classifier using LASSO logistic regression, with 22 proteins that had shown significant associations in the discovery analyses as input variables (Table 3). The resulting models discriminated carriers from noncarriers reasonably well, with area under the curves of 0.752 for *TET2*, 0.898 for *DNMT3A*, and 0.784 for *PPM1D* (Figure S1). A composite CHIP-ProtRS

Table 2. Distribution of CHIP Mutations by Gene According to the Primary End Point

Gene	Primary end point (+; n=33)	Primary end point (−; n=85)	P value
TET2	2 (6.1)	9 (10.6)	0.684
DNMT3A	4 (12.1)	7 (8.2)	0.765
ASXL1	2 (6.1)	5 (5.9)	0.999
PPM1D	2 (6.1)	7 (8.2)	0.990
TP53	1 (3.0)	3 (3.5)	0.999
SF3B1	2 (6.1)	1 (1.2)	0.389
SRSF2	0 (0.0)	2 (2.4)	0.925
JAK2	0 (0.0)	1 (1.2)	0.999
KIT	0 (0.0)	1 (1.2)	0.999

Data are presented as number (%). CHIP indicates clonal hematopoiesis of indeterminate potential.

Table 3. List of Proteins Associated With CHIP Mutation With PCA $P < 0.05$ That Were Used for Generating CHIP-ProtRS in Derivation Cohort

Protein name	UniProt ID	Symbol	Standardized β	Corrected P value
<i>TET2</i>				
Colony-stimulating factor 3 receptor	Q99062	<i>CSF3R</i>	1.342	0.001
Interleukin-4 receptor	P24394	<i>IL4R</i>	1.292	0.003
Cartilage intermediate layer protein 1	O75339	<i>CILP</i>	1.251	0.006
Nuclear receptor ROR-gamma	P51449	<i>RORC</i>	1.218	0.009
Peptidylprolyl <i>cis-trans</i> isomerase B	P23284	<i>PIIB</i>	1.216	0.009
Interleukin-1 beta	P01584	<i>IL1B</i>	1.127	0.031
C-C motif chemokine 18	P55774	<i>CCL18</i>	1.097	0.047
<i>DNMT3A</i>				
Beta-defensin 135	Q30KP9	<i>DEFB135</i>	1.539	<0.001
Potassium voltage-gated channel subfamily E member 3	Q9Y6H6	<i>KCNE3</i>	1.278	0.004
Cadherin-6	P55285	<i>CDH6</i>	1.266	0.005
PDZ domain-containing protein 7	Q9H5P4	<i>PDZD7</i>	1.262	0.005
Homocysteine-responsive ER-resident ubiquitin-like domain member 2	Q9BSE4	<i>HERPUD2</i>	1.198	0.012
Ly6/PLAUR domain-containing protein 3	O95274	<i>LYPD3</i>	1.177	0.016
Protein ATP1B4	Q9UN42	<i>ATP1B4</i>	1.138	0.028
Complement C1s subcomponent	P09871	<i>C1S</i>	-1.118	0.034
Integrin beta-2	P05107	<i>ITGB2</i>	1.104	0.043
Bleomycin hydrolase	Q13867	<i>BLMH</i>	1.092	0.050
<i>PPM1D</i>				
Succinate-semialdehyde dehydrogenase, mitochondrial	P51649	<i>ALDH5A1</i>	-1.452	0.002
FAD-linked sulfhydryl oxidase ALR	P55789	<i>GFER</i>	1.374	0.006
Synaptotagmin-2	Q8N9I0	<i>SYT2</i>	1.303	0.016
SUN domain-containing protein 3	Q8TAQ9	<i>SUN3</i>	1.258	0.022
T-cell immunomodulatory protein	Q8TB96	<i>ITFG1</i>	1.246	0.031

ALR indicates augments of liver regeneration; C1s, complement component 1 subcomponents; CHIP-ProtRS, clonal hematopoiesis of indeterminate potential proteomic risk score; ER, endoplasmic reticulum; FAD, flavin adenine dinucleotide; PDZ, PSD95-Dlg1-ZO1 domain; PLAUR, plasminogen activator urokinase receptor; ROR, retinoic acid-related orphan receptor; and SUN, Sad1, and UNC84 domain-containing.

was then constructed from the 3 gene-specific scores using a LASSO Cox model. In this model, only the *TET2*-derived score received a nonzero coefficient, indicating that the *TET2* proteomic signature was the primary driver of prognostic discrimination. Patients with primary outcome events exhibited higher CHIP-ProtRSs compared with those without (0.56 [IQR, 0.37–0.64] versus 0.44 [IQR, 0.32–0.59]; $P=0.034$).

Performance of the CHIP-ProtRS for Estimating Clinical Outcomes

In the derivation cohort, 33 (28.0%) participants experienced the primary end point, including 19 cardiac deaths (16.1%). Secondary outcomes included 28 all-cause deaths (23.7%), 5 myocardial infarctions (4.2%), and 4 strokes (3.4%). The CHIP-ProtRS significantly stratified the risk of the primary end point (log-rank $P=0.047$; Figure 4A). A total of 227 of 654 (34.7%) participants in the external validation cohorts reached the primary end point, and 125 (19.1%) experienced all-cause mortality

during follow-up. The CHIP-ProtRS again showed significant risk stratification (log-rank $P=0.001$; Figure 4B).

In the derivation cohort, the CHIP-ProtRS remained an independent predictor of the primary end point after adjustment for age, sex, MAGGIC risk score, NT-proBNP, and AF (adjusted HR, 1.42 [95% CI, 1.01–2.00]; $P=0.041$). Similarly, in the external validation cohorts, the CHIP-ProtRS showed an independent association with the primary end point after the same covariate adjustment (HR, 1.16 [95% CI, 1.01–1.32]; $P=0.034$). For secondary outcomes, the CHIP-ProtRS independently predicted major adverse cardiovascular events in the derivation cohort (adjusted HR, 1.52 [95% CI, 1.11–2.07]; $P=0.009$). In the external validation cohorts, participants with higher CHIP-ProtRSs also had a greater risk of all-cause mortality (adjusted HR, 1.29 [95% CI, 1.07–1.54]; $P=0.007$; Table 4). Adding the CHIP-ProtRS to the baseline clinical model improved model fit in both the derivation (likelihood ratio $\chi^2=4.15$; $P=0.042$) and external validation cohorts ($\chi^2=4.47$; $P=0.035$). However, the

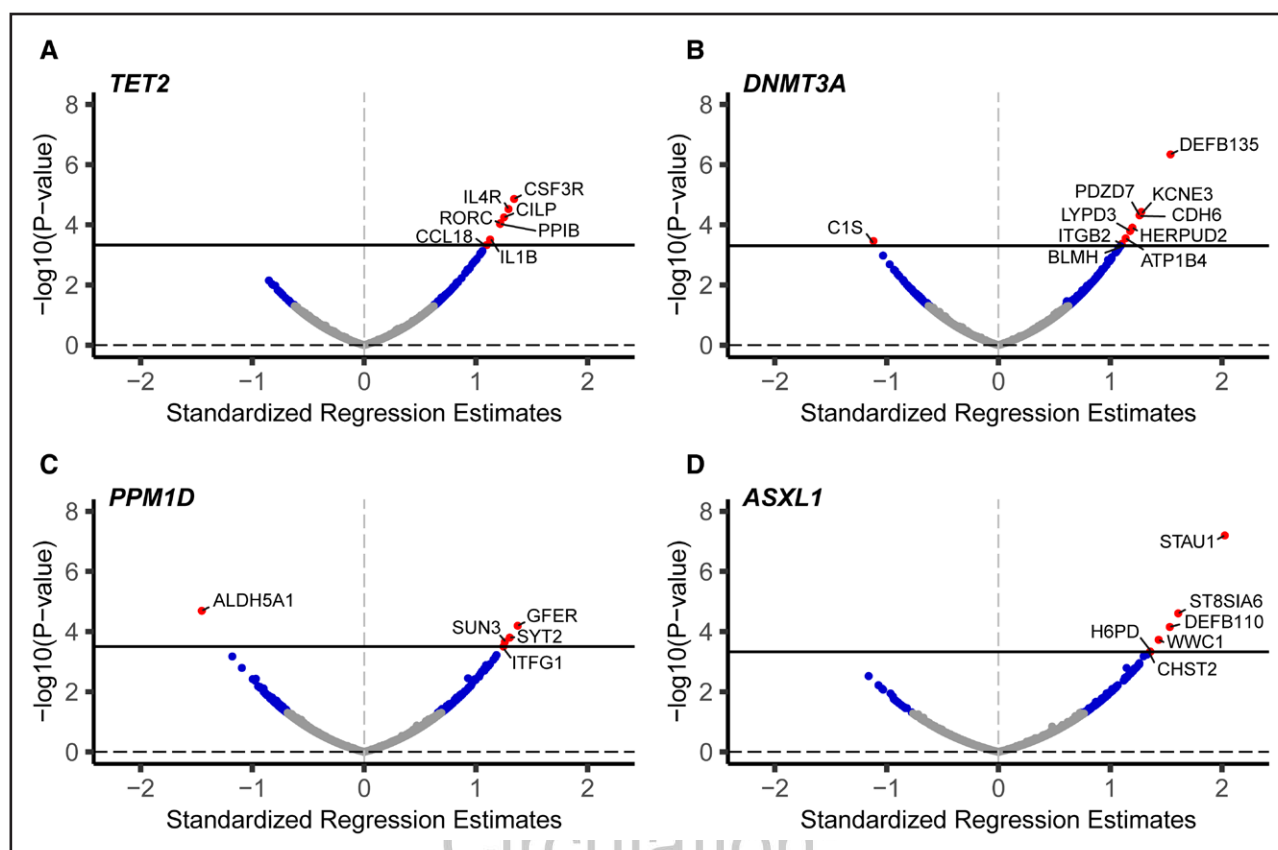


Figure 2. Volcano plots showing proteins significantly associated with clonal hematopoiesis of indeterminate potential-associated genes: *TET2* (A), *DNMT3A* (B), *ASXL1* (C), and *PPM1D* (D) in the TOPCAT cohort (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist).

corresponding changes in discrimination and reclassification, assessed by the C-index, integrated discrimination improvement, and continuous net reclassification improvement, were not statistically significant (Table S3).

Sensitivity Analyses

Several sensitivity analyses were performed to evaluate the robustness of the association between CHIP-ProtRS and the primary composite end point (Table S4). When CHIP carrier status (variant allele frequency $\geq 1\%$) was additionally included as a covariate in the multivariable Cox model, the CHIP-ProtRS remained independently associated with the primary end point (adjusted HR per SD, 1.43 [95% CI, 1.01–2.01]; $P=0.041$), indicating that the score captures prognostic information beyond binary CHIP mutation status. The association was similarly robust after additional adjustment for gene-specific mutation status (*DNMT3A*, *TET2*, and *PPM1D* individually), body mass index, and after exclusion of 3 patients harboring comutations among the 3 modeled genes (all $P<0.05$). To assess the influence of covariate selection in the protein-mutation association step, the entire score derivation pipeline—including protein-mutation

association, gene-specific LASSO classifiers, and Cox LASSO combination—was repeated, substituting sex, spironolactone treatment, body mass index, or estimated glomerular filtration rate for age in the linear regression models. In all 4 models, the reconstructed CHIP-ProtRS remained significantly associated with the primary end point (adjusted HR per SD ranging from 1.44 to 1.60; all $P<0.05$). These results indicate that the prognostic association of CHIP-ProtRS is robust across multiple analytic specifications.

DISCUSSION

In this multicohort study of participants with HFpEF, we identified distinct circulating protein signatures associated with major CHIP driver gene mutations and demonstrated their prognostic relevance. Gene-specific proteomic classifiers for *TET2*, *DNMT3A*, and *PPM1D* showed strong discrimination in the derivation cohort; however, in the Cox-LASSO combination, only the *TET2*-derived score retained a nonzero coefficient, indicating that *TET2*-associated proteomic alterations are the dominant prognostic contributor among CHIP-related signals in HFpEF. The resulting CHIP-ProtRS robustly stratified the risk of death and HF

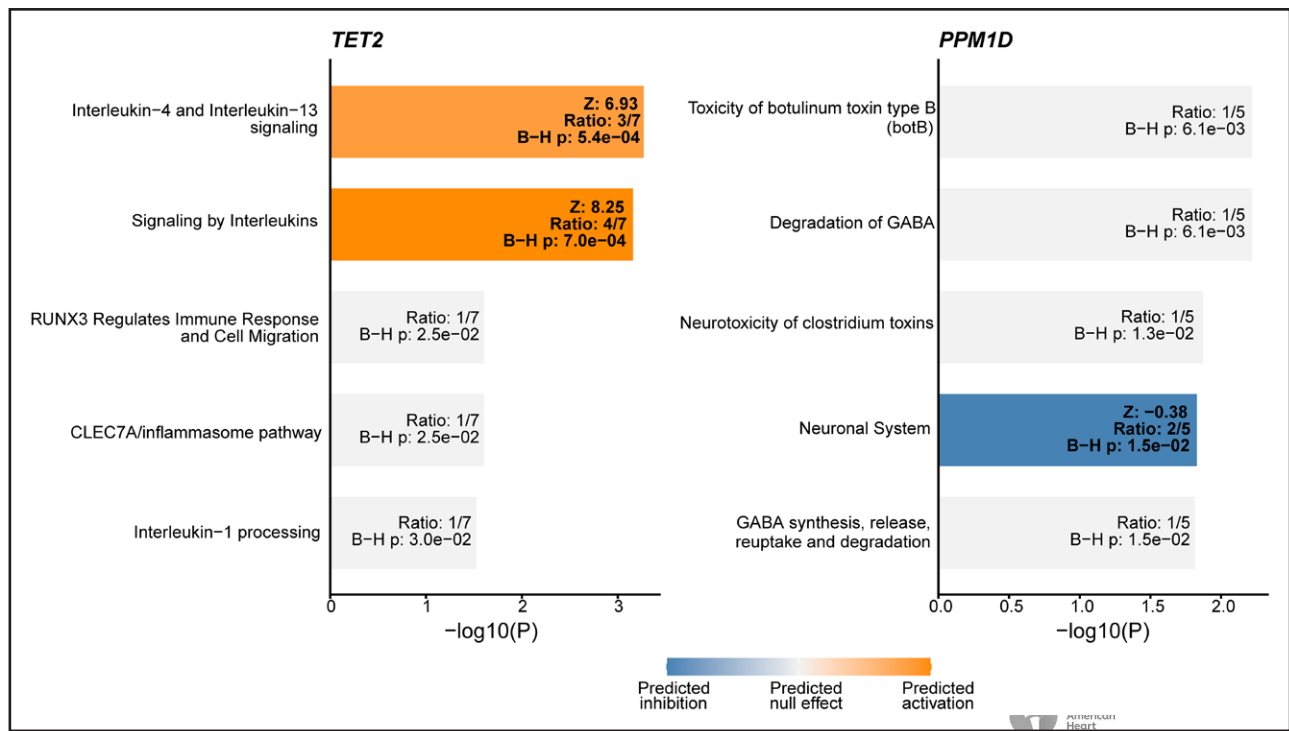


Figure 3. Pathways significantly correlated with the clonal hematopoiesis of indeterminate potential-associated genes in the TOPCAT cohort (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist): *TET2* and *PPM1D*. All displayed pathways met the false discovery rate <0.05 threshold for significance. The top 5 enriched pathways per gene, ranked by adjusted *P* value, are displayed. Orange bars indicate predicted pathway activation (positive *Z* score), blue bars indicate predicted inhibition (negative *Z* score), and gray bars represent pathways for which directionality could not be assessed (fewer than 2 input genes mapped). *Z* scores were calculated by aggregating gene-level *Z* statistics within each pathway. B-H *p* indicates Benjamini-Hochberg-adjusted *P* value.

hospitalization. Importantly, the CHIP-ProTRS remained an independent predictor of adverse outcomes after adjustment for age, sex, AF, NT-proBNP, and MAG-GIC risk score, with consistent performance across the external validation cohorts. To our knowledge, this is the first study to use large-scale proteomic profiling to characterize CHIP-related abnormalities and to demonstrate its association with clinical outcomes in patients with HFpEF.

Increasing evidence suggests that CHIP may contribute to the development of HFpEF.^{7–9} Population-based analyses further support this association by demonstrating an increased risk of incident HFpEF among individuals with CHIP, implicating mutant hematopoietic clones in diastolic dysfunction.⁹ In addition, among patients with established HF, the presence of CHIP, especially *DNMT3A* or *TET2*, has been associated with accelerated HF progression.²⁴ These clinical observations are supported by experimental data: loss of *Tet2* function promotes the expression of IL-1β, IL-6, and Ccl5, leading to myocardial fibrosis.²⁵ Moreover, *Tet2* mutations drive the expansion of Spp1⁺ macrophages, which may promote myocardial fibrosis in HFpEF.⁷ In mouse models with *Dnmt3a* inactivation, macrophage stimulation results in upregulation of Cxcl1, Cxcl2, IL-6, and Ccl5, causing cardiac hypertrophy, impaired cardiac function,

and myocardial fibrosis.²⁵ Single-cell level analyses of patients with HF demonstrate that expansion of *DNMT3A* mutant clones activates wild-type macrophages and CD4⁺ T cells, which may indirectly promote cardiomyocyte hypertrophy and fibrosis via paracrine mechanisms²⁶ and enhance transcriptional signatures related to monocyte-mediated inflammation and T-cell activation.²⁷ In CHIP, *PPM1D* mutations impair DNA damage repair by acting as gain-of-function variants, leading to the accumulation of DNA damage.²⁸ *PPM1D* mutations increase reactive oxygen species and, in turn, elevate IL-1β and IL-18 in macrophages.²⁹ Transplantation of *PPM1D*-mutant hematopoietic cells into mice resulted in markedly aggravated left ventricular hypertrophy after angiotensin II infusion, a phenotype that can contribute to HFpEF development.²⁹ Consistent with this biology, our proteomic analyses identified circulating protein patterns associated with these mutations.

The biological roles of the proteins associated with CHIP in our analysis provide mechanistic clues regarding the underlying biological abnormalities in patients with HFpEF with CHIP. CILP1 (cartilage intermediate layer protein 1), produced by ventricular fibroblasts, is enriched in myocardial tissue from patients with myocardial infarction or severe aortic stenosis and contributes to fibrosis-related processes, including collagen

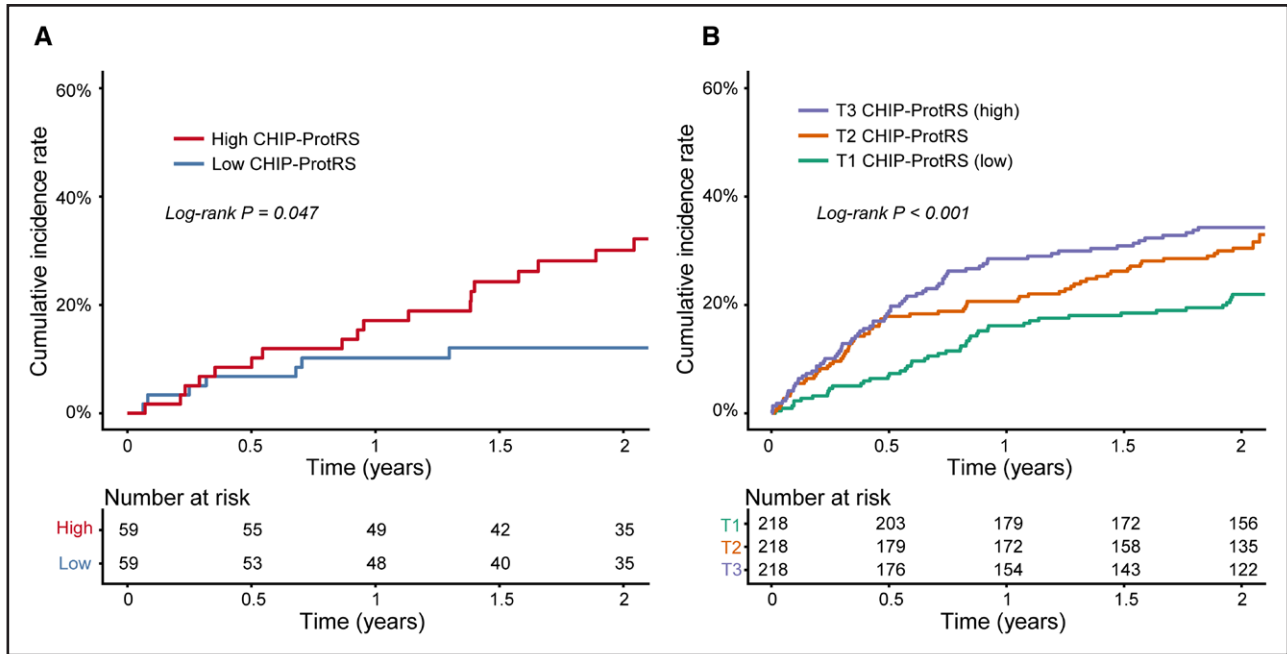


Figure 4. Cumulative incidence curves of the primary end point by clonal hematopoiesis of indeterminate potential proteomic risk score (CHIP-ProtRS) tertiles in the TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) (A) and external validation cohorts (B).

degradation and extracellular matrix organization.³⁰ Cilp1 expression was shown to promote myofibroblast proliferation and myocardial fibrosis in murine myocardial infarction models.³¹ The positive association between *TET2* mutations and CILP1 in our HFpEF cohort—despite the absence of myocardial infarction—suggests a potentially relevant link that warrants further investigation. RORC (nuclear receptor ROR- γ), a key transcription factor in Th17 cell differentiation, induces IL-17A production, and IL-17A has been shown to drive myocardial fibrosis through activation of the PKC β /Erk1/2/NF- κ B pathway.³² An additional study indicates that IL-17A enhances cardiac expression of IL-6, TNF α , and IL-1 β and promotes the recruitment of CD11b(+) monocytes and Gr1(+) granulocytes, in turn contributing to myocardial fibrosis.³³ CCL18 (C-C motif chemokine 18), mainly produced by M2 macrophages, is highly expressed in chronic inflammation and may drive lymphocyte recruitment and collagen deposition by fibroblasts.³⁴ In the Heart and Soul Study, plasma CCL18 predicted future HF hospitalization and was associated with worse renal function in participants with chronic kidney disease.³⁵ Among the *TET2*-associated proteins, IL-1 β is the most extensively characterized mediator of CHIP-related cardiovascular inflammation, consistent with the well-established *TET2*-NLRP3 (NOD-like receptor pyrin domain-containing protein 3)-IL-1 β signaling axis.^{36,37} CCL18, a monocyte/macrophage-derived chemokine

Table 4. Multivariable Cox Regression Analysis of the CHIP-ProtRS for the Primary and Secondary End Point in the Derivation Cohort and the External Validation Cohorts

Outcomes	Event/participants	HR* per SD (95% CI)	P value
Derivation cohort			
Primary end point (cardiac death or HF hospitalization)	33/118	1.42 (1.01–2.00)	0.041
Cardiac death	19/118	1.41 (0.90–2.22)	0.130
HF hospitalization	21/118	1.49 (0.98–2.28)	0.062
MACE	38/118	1.52 (1.11–2.07)	0.009
External validation cohort			
Primary end point (all-cause death or HF hospitalization)	227/654	1.16 (1.01–1.32)	0.034
All-cause death	125/654	1.29 (1.07–1.54)	0.007

CHIP-ProtRS indicates clonal hematopoiesis of indeterminate potential proteomic risk score; HF, heart failure; HR, hazard ratio; MACE, major adverse cardiovascular events; MAGGIC, Meta-Analysis Global Group in Chronic Heart Failure; and NT-proBNP, N-terminal pro-B-type natriuretic peptide.
*Adjusted with age, sex, NT-proBNP, MAGGIC risk score, and atrial fibrillation.

linked to inflammatory cell recruitment and collagen deposition.^{34,35} In contrast, several proteins identified in our analysis—including CSF3R (colony-stimulating factor 3 receptor), IL4R (interleukin-4 receptor), RORC, CILP, and PPIB (peptidylprolyl isomerase B)—represent novel associations not previously reported in the CHIP literature, suggesting previously unrecognized pathways through which *TET2*-mutant CH may contribute to HFpEF pathobiology. KCNE3 (potassium voltage-gated channel subfamily E member 3), a regulator of several potassium channels, has been implicated as a potential causal contributor to AF.³⁸ *Kcne3*-deficient mice exhibit a 2.3-fold increase in serum aldosterone, and KCNE3 deletion induces ventricular arrhythmogenicity through secondary hyperaldosteronism and adrenal lymphocyte infiltration, which impair ventricular repolarization.³⁹ HERPUD2 (homocysteine-responsive ER-resident ubiquitin-like domain member 2), an endoplasmic reticulum-localized stress-response protein, has been linked to hyperuricemia in exome-wide association studies.⁴⁰ Previous studies have attributed the association between CHIP and gout primarily to *TET2*-IL-1 β signaling.⁴¹ Our findings, however, raise the possibility that *DNMT3A*-related HERPUD2 upregulation and hyperuricemia may also underlie this relationship. ALDH5A1 (succinate-semialdehyde dehydrogenase) and GFER (FAD-linked sulfhydryl oxidase ALR) are mitochondrial proteins, and mitochondrial dysfunction is known to contribute to abnormal myocardial stiffness, skeletal muscle function, and HF progression by disrupting energy metabolism and redox balance.⁴²

Given that the final CHIP-ProtRS is derived from the *TET2*-associated proteomic signature, an important question for clinical translation is whether it provides information beyond targeted assessment of *TET2* CH, which is increasingly accessible in clinical practice. Although prior larger studies have linked *TET2* CHIP to adverse outcomes in HF,^{7,9} the prevalence of *TET2* mutations did not differ significantly between participants with and without the primary end point in our modestly sized derivation cohort. Beyond considerations of statistical power, mutation status and variant allele frequency primarily capture the presence and burden of CH but do not fully reflect whether the clone is biologically active or the extent to which it is modulating downstream inflammatory, immune, or cardiovascular pathways. By reflecting downstream inflammatory and immune alterations that vary with clone size, environmental exposures, and concurrent therapies, CHIP-ProtRS may therefore capture aspects of pathobiologically active CHIP that are not reflected by mutation-based assessment alone. Thus, rather than serving as a replacement for established clinical risk-prediction tools or CHIP sequencing, the CHIP-ProtRS may provide a complementary functional readout of downstream CHIP-related biological activity, with potential applications in biological phenotyping and,

if validated prospectively, enrichment of trials targeting CHIP-related inflammatory pathways.

Strengths and Limitations

This study has several strengths. First, unlike prior work that evaluated prognosis solely based on the presence or absence of CHIP, we utilized CHIP-related alterations in circulating protein levels. We demonstrated their independent association with clinical outcomes in HFpEF. Second, the consistent prognostic association of the CHIP-ProtRS across multiple independent cohorts supports the robustness and generalizability of our findings. Third, we showed that CHIP status can be inferred from proteomic profiling, enabling identification of individuals with CHIP-like biological activity even in the absence of sequencing data. Fourth, all cohorts were profiled using the same proteomic platform, minimizing measurement heterogeneity and strengthening cross-cohort comparability. In addition, the CHIP-associated proteins identified in our analysis highlight biologically relevant pathways that may serve as potential therapeutic targets in HFpEF. However, further studies are needed to determine their causal roles and modifiability.

Nevertheless, several limitations should be acknowledged. The derivation cohort had a modest sample size, limiting our ability to characterize proteomic signatures associated with less common CHIP driver genes previously linked to HF, such as *ASXL1* and *TP53*. In addition, because the subset of TOPCAT participants with available blood samples was drawn exclusively from the American study sites, generalizability to broader HFpEF populations may be limited. To address this, we externally validated the proteomic classifiers in 3 independent HFpEF cohorts (including a cohort from New Zealand) with distinct demographic and clinical profiles, demonstrating consistent associations across heterogeneous populations. Third, external validation cohorts lacked CHIP mutation data, preventing direct evaluation of classifier discrimination outside the derivation cohort. However, the distinctly learned proteomic signatures associated with CHIP status carried independent information relevant to clinical outcomes in the external validation cohorts. In addition, given the observational nature of the study, residual confounding cannot be excluded despite multivariable adjustment. Fourth, the primary composite end point differed between the derivation (cardiovascular death or HF admission) and validation (all-cause death or HF admission) cohorts, reflecting differences in available outcome adjudication; however, the use of all-cause death in the validation cohort constitutes a more conservative test that would dilute any CHIP-specific signal toward the null. Fifth, unlike somatic mutations, which are relatively stable genetic alterations, circulating protein levels may be influenced by concurrent comorbidities, treatment effects, and other physiological changes; the



temporal stability of CHIP-ProtRS could not be assessed in the absence of serial proteomic measurements. Finally, large-scale proteomic profiling is not yet available in routine clinical practice, which currently limits the direct clinical applicability of the CHIP-ProtRS.

Conclusions

We identified associations between CHIP and circulating proteins in patients with HFpEF. Using these proteomic signatures, we developed a CHIP-ProtRS and confirmed its independent association with adverse clinical outcomes in independent external cohorts. Whether the proteins linked to CHIP have the potential to serve as therapeutic targets for CHIP-directed interventions in HFpEF warrants further investigation.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S4

Figure S1

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